

API Monitoring & Responsible Use Plan

Deploying the Churn Prediction API is only the first step. To ensure the model remains reliable, accurate, and fair in production, the following monitoring and usage protocols must be adhered to.

1. Monitoring Plan

A. Data Drift Tracking

- **What to track:** The statistical distribution of incoming feature payloads over a 30-day rolling window compared to the training data.
- **Key metrics:** Average recency_days, sessions_30d, and shift in acquisition_channel distributions.
- **Alert Trigger:** If the Population Stability Index (PSI) for any key continuous variable exceeds 0.2, trigger a data audit.

B. Prediction Distribution Tracking

- **What to track:** The percentage of customers being classified as predicted_class: 1 (Churn) vs. 0 (Retain).
- **Alert Trigger:** If the API suddenly flags > 40% of the customer base as "High Risk", this indicates either an unprecedented business crisis or an upstream feature bug (e.g., website session tracking is broken, causing all sessions_30d to equal 0).

C. Business Outcomes Tracking

- **What to track:** The actual 60-day churn rate of the cohorts flagged by the API.
- **Evaluation Metric:** Compare the predicted probability against the observed conversion rate (Calibration). Calculate monthly F1-score on a delayed basis (after 60 days).

D. API Errors & System Health

- **What to track:** * HTTP 422 Unprocessable Entity: Spikes indicate the CRM system is sending incorrectly formatted data.
 - HTTP 500 Internal Server Error: Tracks pipeline or Pandas conversion failures.
 - *Latency:* Predict endpoint should respond in < 150ms to prevent CRM UI blocking.

E. Retraining Triggers

The model must be retrained manually or automatically if:

1. The F1-Score on recent historical data drops below 0.60.
2. A new feature column is added to the CRM (e.g., a new Loyalty Program tier).
3. 6 months have passed since the last training (model.pkl timestamp).

2. Responsible Use Guidelines

This API is designed to **assist** the retention team, not replace human judgment.

✓ HOW the output SHOULD be used:

- **Service Recovery Prioritization:** If a customer is flagged as "High Risk" and the risk_explanation indicates a "negative support experience," customer service agents should bump this user to the top of the queue for VIP outreach.
- **Targeted Incentives:** Use the risk_level to determine the aggressiveness of retention offers. Low-risk customers should not receive heavy discounts, protecting profit margins.

✗ HOW the output SHOULD NOT be used:

- **Denial of Service/Punishment:** The churn probability must **never** be used to intentionally degrade a customer's experience (e.g., routing them to a slower support queue because they are "going to leave anyway").
- **Automated Discount Spamming:** Automatically emailing a 50% discount to everyone the model flags as "High Risk" will train customers to feign inactivity to trigger the system. Always pair the prediction with the risk_explanation to choose the correct, non-monetary intervention first.